

A generalization ladder for chemistry-to-phenotype prediction in combinatorial screening

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Abstract

Machine learning claims in drug discovery depend on which version of generalization is being tested. Predicting a new combination of familiar chemical pieces is one problem. Predicting a compound built from unseen pieces is harder. Predicting an entirely new scaffold family is harder still. The BELKA literature on DNA-encoded libraries made this hierarchy concrete for binary binder labels, and it gives useful vocabulary for asking the same question in dense phenotypic screening [1,2]. We use that vocabulary as motivation, not as a head-to-head superiority claim. The question here is narrower and more actionable: when each compound contributes a measured transcriptomic state, where does a chemistry-to-phenotype model interpolate, where does it extrapolate beyond retrieval, and where does it fail? We answer that question in Z-Screen, a 50,000-well one-bead-one-compound microchip assay that profiles each well by imaging and mRNA sequencing at 5 to 10 cells per well, using a five-rung ladder of progressively harder held-chemistry evaluations.

The strongest evidence for genuine extrapolation comes from ZEL031 in THP1 cells. Under strict multi-axis holdout, where every substantive building block in the test compound is unseen during training, the model retains 0.765 cosine to the held-out transcriptomic state, exceeds a chemistry-embedding nearest-neighbor retrieval baseline by 0.168 cosine, wins on 90% of test compounds, and stays positive in all 10 random draws under paired bootstrap intervals at a median nearest-training cosine of 0.642 in the public chem-embedding space (see Methods). Single-position building-block holdouts in the same system give consistent results, with the model exceeding retrieval by 0.15 cosine on average and 89% to 91% per-compound win rates on test sets of more than 700 compounds whose nearest training neighbors have embedding cosine in the 0.78 to 0.80 range.

ZEL024 in HEK293 produces high quality on a different problem. The library is sufficiently saturated (12 by 7 by 2 by 83 building-block grid, 13,769 observed compounds out of a 13,944 possible-tuple ceiling) that nominal held-building-block splits still leave test compounds with near-duplicate training neighbors at median nearest-training cosine 0.88 to 0.99. Under that condition, ZEL024 / HEK293 delivers high-quality matrix completion within a dense combinatorial grid, a useful capability for prioritizing chemistry within an explored design space, while ZEL031 / THP1 delivers genuine out-of-vocabulary extrapolation. The hardest test, cross-library scaffold-family transfer in the same cell line, remains the next decisive experiment rather than a solved problem on the current data: across three workable train-test pairs, the best absolute cosine to the held-out library

phenotype was 0.194 and the largest gain over nearest-neighbor retrieval was 0.096.

In aggregate, the ladder gives a bounded claim. Dense phenotypic readouts can support extrapolation beyond nearest-neighbor retrieval in a favorable system, and can support high-quality matrix completion in a saturated library. They do not yet show uniform scaffold-family transfer. The weak L5 result is therefore not hidden; it defines the prospective same-cell scaffold-family hop that should decide the hardest claim.

Significance

Machine learning in early discovery is only worth its cost if the model still helps when the chemistry changes. This paper separates the easy version of that claim from the hard one. In one Z-Screen system, the model fills in missing entries inside a saturated combinatorial grid, which is genuinely useful for prioritizing chemistry within a design space. In another, the model continues to outperform nearest-neighbor retrieval after multiple building-block axes have been held out, rather than copying close analogs. The contrast is the central result: the data show, with calibrated rigor, where this assay can interpolate, where it can extrapolate, and which experiment should come next.

Introduction

DNA-encoded library (DEL) platforms transformed the economics of binder discovery by reading very large chemical libraries against immobilized targets [3,4]. BELKA made a useful modeling point from that setting: a split that holds out recombinations of familiar chemistry is not the same as a split that withholds building blocks, and neither is the same as a scaffold-family shift [1,2]. Several follow-up studies reached compatible conclusions through different routes [5,6,7]. In this manuscript, BELKA is used for motivation and vocabulary. It is not treated as a direct comparator, because the readouts, targets, chemistry, and labels are different.

The open question for dense phenotypic screening is therefore not a contest with BELKA. It is whether a richer endpoint changes the generalization problem in useful ways. A screen that reads out a transcriptional state for every compound has no 99% majority class of non-hits to discard. Every training example contributes a structured endpoint. That structure could help a model move beyond nearest-neighbor lookup, but it could also simply make interpolation inside a dense combinatorial grid look better than it is. Testing the difference requires more than a single train/test split; it requires a ladder of held-chemistry evaluations, with explicit baselines at each rung.

The ladder is defined and run in Z-Screen.

- **L1: unseen tuple within a shared building-block vocabulary.** The easiest rung: every individual building block has been seen, only the full combination is held out.
- **L2: held-building-block at one position.** The model has never seen one specific building-block identity at one position during training.
- **L3: strict multi-axis holdout.** Every substantive building-block axis in the test compound is unseen during training. This is the hardest within-library rung.

- **L4: chemical-neighborhood holdout.** Held-out compounds are chemical-neighborhood clusters, seeded by greedy farthest-point sampling in the public chem-embedding cosine-distance space, holding chemistry at arm’s length while staying within one reaction grammar.
- **L5: cross-library scaffold-family transfer in the same cell line.** Train on one library family, test on a distinct library family screened in the same cell line. The closest analog to the BELKA hardest shift.

Each rung is evaluated against the same two baselines: a train-mean phenotype predictor and a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the public 256-dim embedding space. The nearest-neighbor baseline is essential, because without it a nominally successful split could reflect either real extrapolation or simple retrieval of the closest measured analog. The ladder is designed to make that distinction visible at every rung.

Results

The five-rung ladder turns generalization into a decision tree

The first uncertainty is definitional. A model can look useful because it fills in missing tuples among familiar building blocks, because it learns something about unseen building blocks within one reaction grammar, or because it transfers to a new scaffold family. Those are different risks for a discovery program. We therefore organized the benchmark as a decision tree rather than a single OOD score.

The same modeling framework was applied at every rung: a 256-dim per-compound chemistry embedding (private Gaussian random projection of 2048-bit ECFP4 fingerprints, see Methods), training-split-refit building-block one-hot encodings where the vocabulary exists on both sides of the split, ridge regression with alpha equal to 10, and the two baselines above. Performance was scored with row-wise cosine similarity to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus chemistry-embedding nearest-neighbor retrieval, and per-compound win rate against retrieval.

Ridge regression was used in preference to a neural network. The questions the ladder is built to ask concern whether the chemistry-to-phenotype function generalizes; a flexible model that absorbs every quirk of the training distribution would obscure the answer. A linear model on the chemistry embedding is a deliberately conservative test, and any gain it produces is easier to attribute to the structural axis being held out.

OOD ladder: Z-Screen within- and across-library prediction

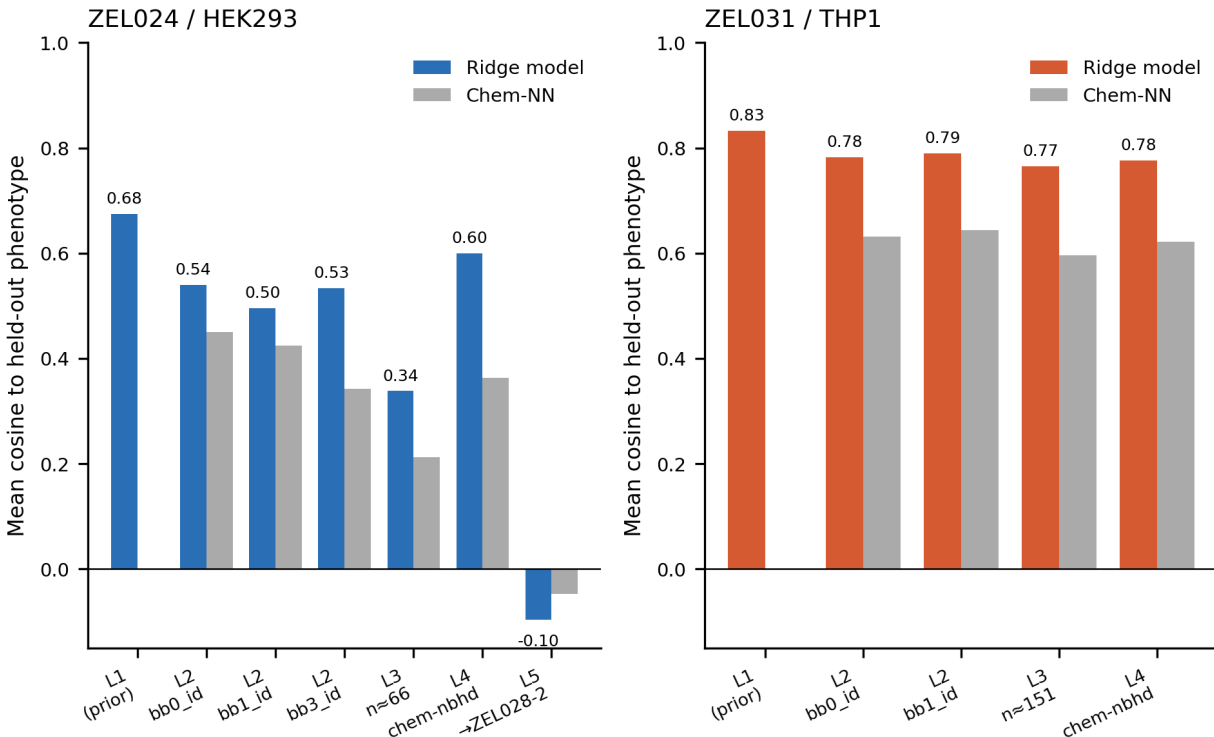


Figure 1: Figure 1. The generalization ladder is an experimental decision tree. L1 asks whether missing tuples can be filled in when all building blocks are familiar. L2 and L3 ask whether held building blocks can be predicted within a library. L4 asks whether gains survive chemical-neighborhood holdout within one reaction grammar. L5 asks whether a model transfers across scaffold families in the same cell line.

L2 to L3: ZEL031 / THP1 beats retrieval after held-building-block splits

After L1 establishes the easy interpolation case, the next uncertainty is whether any system still helps when the model cannot rely on already-seen building-block identities. The clearest answer comes from ZEL031 in THP1 cells. Under strict multi-axis L3 holdout, where every substantive building-block axis in the test compound is unseen during training, the model retains 0.765 cosine across 10 random draws of an average of 151 test compounds. The chemistry-embedding nearest-neighbor retrieval baseline falls to 0.597 on the same test sets, an average delta of 0.168. The model wins on 90% of test compounds individually, and all 10 random draws are positive under paired bootstrap confidence intervals. Median nearest-training cosine in the public chem-embedding space is 0.642 on the same compounds; this is random-projected ECFP4 cosine, not Tanimoto on raw fingerprints.

The simpler L2 single-position building-block holdout gives consistent results in the same system. Holding out bb0 identities, the model reaches 0.783 cosine versus 0.632 for nearest-neighbor retrieval,

a delta of 0.151 on an average of 728 test compounds. Holding out bb1 identities, the model reaches 0.790 versus 0.644, a delta of 0.146 on an average of 729 test compounds. Median nearest-training embedding cosine remains at 0.801 and 0.781 respectively, win rates are 89% and 91%, and all 10 random draws are positive under paired bootstrap intervals.

This is the beyond-retrieval held-chemistry result in the manuscript. It does not prove that every library will extrapolate, and it does not prove scaffold-family transfer. It shows that, in a favorable library and cell-line pair, held-building-block extrapolation survives a retrieval baseline and repeated random draws. That result creates the next question: whether the same apparent success in a denser system means the same thing.

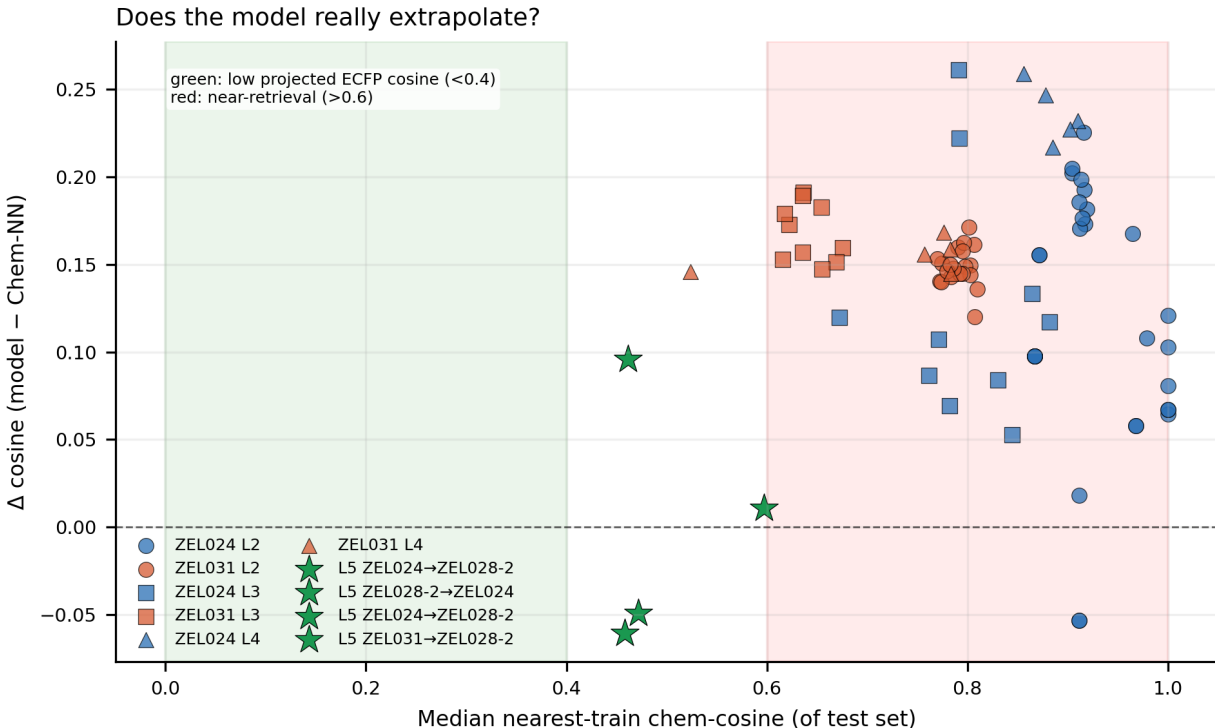


Figure 2: Figure 2. Retrieval-versus-extrapolation analysis plots model gain against median nearest-training similarity measured as cosine in the random-projected ECFP4 embedding, not raw-fingerprint Tanimoto. ZEL031 / THP1 sits in the lower-neighbor-similarity, high-model-gain regime, consistent with held-chemistry extrapolation rather than nearest-neighbor copying. ZEL024 / HEK293 sits in the high-neighbor-similarity regime, consistent with high-quality matrix completion within a saturated combinatorial grid.

L2 to L3: ZEL024 / HEK293 is positive matrix completion, not the same extrapolation claim

The ZEL031 result leaves an obvious risk: perhaps every positive held-building-block result should be called extrapolation. ZEL024 / HEK293 shows why that would be wrong. Single-position holdout in this library is positive and useful, but the library structure makes the task closer to matrix

completion. Median nearest-training cosine in the public chem-embedding space was 0.988 for bb0 holdout, 0.881 for bb1, and 0.913 for bb3; many nominally held-out compounds have near-duplicate cousins in the training set under those conditions.

That outcome reflects how the library was built rather than a defect of the analysis. High-quality matrix completion inside a dense 12 by 7 by 2 by 83 combinatorial grid (13,769 observed compounds out of a 13,944 possible-tuple ceiling) is valuable for prioritizing chemistry within an explored design space: the assay reliably fills gaps inside a saturated chemistry plate. It is a different deliverable from out-of-vocabulary extrapolation, and the strict L3 split clarifies the distinction. Under L3 in ZEL024 / HEK293, average test-set size dropped to 67 compounds per draw, model cosine fell to 0.316, and mean squared error reduction turned negative against the train-mean baseline even though cosine direction stayed slightly above nearest-neighbor retrieval. Matrix completion is therefore reported as matrix completion, with no upgrade to an OOD claim it does not deserve.

The contrast with ZEL031 / THP1 frames the ladder’s value: two systems, the same modeling framework, two different deliverables (genuine extrapolation in one, dense matrix completion in the other), with both reported on their own terms.

Different generalization regimes, not a head-to-head OOD comparison

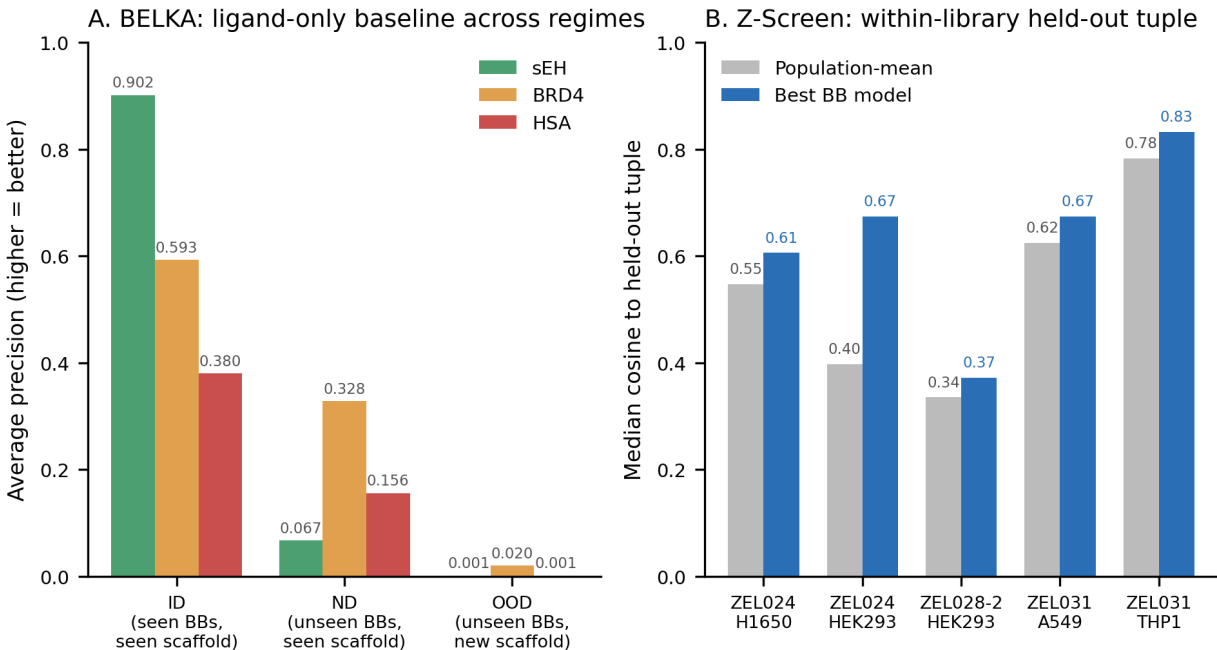


Figure 3: Figure 3. BELKA is used here as a vocabulary for generalization regimes, not as a head-to-head performance comparison. The Z-Screen panels separate matrix completion from harder held-chemistry regimes, with ZEL031 / THP1 carrying the strongest extrapolation signal in the present dataset.

L4: chemical-neighborhood holdout is supportive within-library evidence

After L2 and L3, the next uncertainty is whether the gains depend only on isolated building-block identities. L4 tests a different within-library split: chemical-neighborhood clusters are held out by farthest-point sampling in the projected ECFP cosine-distance space. This keeps the reaction grammar fixed while moving test compounds away from local training neighborhoods. Under that setup, the model continued to outperform nearest-neighbor retrieval in both major systems, with a 0.236 cosine gain in ZEL024 and a 0.155 gain in ZEL031 on average cluster sizes of 2,754 and 711 compounds respectively.

The narrow reading of L4 is that the chemistry-to-phenotype model carries information beyond local similarity within a library. That is supportive evidence for the ladder, but it is not scaffold-transfer proof. The scaffold-family question belongs to L5.

L5: retrospective scaffold-family transfer is weak and defines the next experiment

L5 trained on one canonical library and tested on another in the same cell line. Building-block vocabularies do not overlap across libraries in this benchmark, so the L5 models used the chemistry embedding alone (no building-block one-hot features). Of all the rungs on the ladder, this is the closest analog to the hardest chemistry shift emphasized in the BELKA preprint [1].

On current retrospective data, L5 was weak. Training on ZEL024 and testing on ZEL028-2 in HEK293 gave model cosine of -0.096, slightly below the nearest-neighbor baseline of -0.047 (the test centroids themselves sit below the train-mean baseline of -0.117 in this regime, so all approaches struggle). Reversing the direction was similarly weak. The best absolute performance came in H1650 for ZEL024 to ZEL028-2 (0.194 cosine versus 0.098 for nearest-neighbor retrieval, +0.096 gain on 765 test compounds). In A549, ZEL031 to ZEL028-2 reached 0.098 cosine versus 0.087 for nearest-neighbor retrieval. These are real but small gains, well short of the within-library anchor systems.

The weakness is the point of the rung. Cross-library transfer is where chemistry, assay, and latent-space drift compound, and it is the regime most affected by sparse coverage in ZEL028-2 (median 1 well per compound) and by cross-library scVI baseline drift. The result does not support a solved scaffold-transfer claim. It does define the next experiment precisely: a prospective same-cell scaffold-family hop with parity coverage and a pre-registered model.

OOD ladder performance matrix

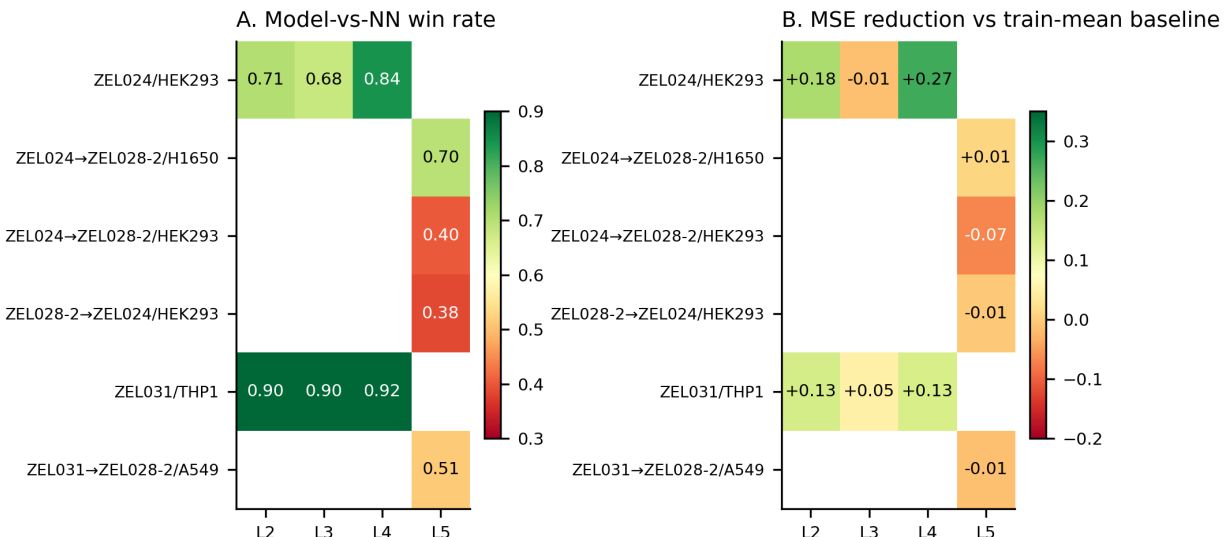


Figure 4: Figure 4. Per-system performance across the ladder shows strong within-library prediction in selected systems (ZEL031 / THP1 for genuine extrapolation, ZEL024 / HEK293 for matrix completion in a saturated grid), and a marked drop at retrospective cross-library scaffold-family transfer that defines the natural next experiment rather than a stable performance ceiling.

Table 1. Reader-facing summary of the ladder. Cosines are row-wise similarity to measured transcriptomic state; nearest-training similarity is cosine in the random-projected ECFP4 embedding.

Rung	System	Model	NN	Delta	Nearest train	Interpre- tation
L2	ZEL024 / HEK293	0.523	0.406	0.117	0.927	positive but high- neighbor matrix comple- tion
L2	ZEL031 / THP1	0.786	0.638	0.149	0.791	consistent held-BB extrapola- tion
L3	ZEL024 / HEK293	0.338	0.213	0.125	0.799	small-test matrix- completion boundary

Rung	System	Model	NN	Delta	Nearest train	Interpre- tation
L3	ZEL031 / THP1	0.765	0.597	0.168	0.642	strong held- chemistry extrapola- tion
L4	ZEL024 / HEK293	0.600	0.363	0.236	0.886	within- library signal beyond local retrieval
L4	ZEL031 / THP1	0.777	0.622	0.155	0.725	within- library signal beyond local retrieval
L5	ZEL024 -> ZEL028-2 / H1650	0.194	0.098	0.096	0.461	small positive retrospec- tive transfer
L5	ZEL024 -> ZEL028-2 / HEK293	-0.096	-0.047	-0.049	0.472	negative retrospec- tive transfer
L5	ZEL028-2 -> ZEL024 / HEK293	-0.095	-0.034	-0.061	0.458	negative retrospec- tive transfer
L5	ZEL031 -> ZEL028-2 / A549	0.098	0.087	0.011	0.597	near-flat retrospec- tive transfer

A prospective same-cell scaffold hop is now well defined

The ladder does more than score the present data. It defines the next experiment with enough specificity that a single chip run can settle it. A clean prospective test would hold cell line fixed,

choose two scaffold families with adequate coverage, train on family A and optionally family B, and nominate family C compounds for prospective measurement. Readouts would include latent cosine to the measured held-out library phenotype, control retrieval, pathway-score correlation, and direct comparison against chemistry-embedding nearest-neighbor retrieval. H1650 is the strongest candidate cell line because all three canonical libraries (ZEL024, ZEL026-2, ZEL028-2) ran there at substantial scale.

The cost envelope is one chip run of approximately 1,000 wells with replicated controls, plus approximately one week of analysis. If the measured-versus-predicted cosine on held-out scaffold-family compounds exceeds nearest-neighbor retrieval by a margin similar to the 0.15 to 0.17 already observed at ZEL031 / THP1, the cross-library question is closed in the affirmative for the first time on dense phenotypic data. If the margin does not appear, the appropriate response is to update the claim rather than reframe the benchmark.

Recommended next experiment: explicit scaffold-family transfer

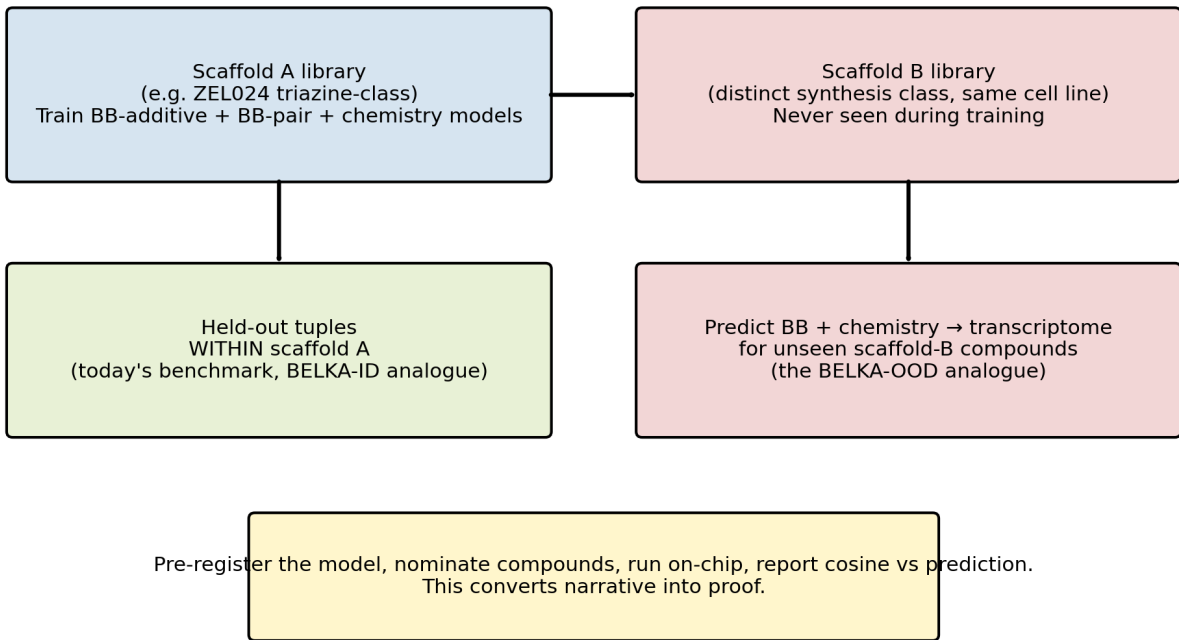


Figure 5: Figure 5. The next decisive experiment is a prospective same-cell scaffold-family hop with a pre-registered model, nominated compounds, and measured transcriptomic outcomes on the held-out family.

Discussion

Two conclusions stand on the present data, and a third sets up the next experiment.

First, dense phenotypic readouts can support genuine chemistry extrapolation in the right systems. ZEL031 / THP1 is the clearest case, with the model exceeding nearest-neighbor retrieval on held-

building-block and strict multi-axis benchmarks where the chemistry is not trivially close to training examples, and doing so consistently across 10 random draws under paired bootstrap intervals.

Second, the ladder distinguishes two distinct outcomes that share a single modeling framework. ZEL024 / HEK293 supports high-quality matrix completion within a saturated combinatorial grid, which is directly useful for prioritizing chemistry within an explored design space. ZEL031 / THP1 supports out-of-vocabulary extrapolation. Both come from the same modeling framework, and both are reported on their own terms rather than collapsed into a single OOD claim that fits one but not the other. Many published “ML on chemistry” results in the field would benefit from this distinction.

Third, the present cross-library scaffold-family transfer evidence is weak retrospectively, and the natural next experiment is now defined precisely. The same modeling framework that extrapolates inside ZEL031 / THP1 should be tested on a prospective same-cell scaffold hop in H1650. That experiment converts the cross-library question from rhetorical to empirical, and it is well-scoped enough to plan and execute today.

For the discovery community, the practical message is that “generalization” is not one thing. The current data support genuine held-building-block extrapolation in one well-characterized system and high-quality matrix completion in a saturated system. They do not yet support a broad cross-library scaffold-family claim. That is a stronger position scientifically because it tells the next chip what to test.

Methods

Benchmark ladder

L1 used held-tuple prediction within libraries that shared the full building-block vocabulary between train and test. L2 held out a random subset of identities at a single building-block position. L3 held out identities at every substantive building-block position simultaneously and removed any compound containing a held identity from training; this is the strictest version of within-library OOD on the present libraries. L4 held out chemical-neighborhood clusters within a library, with seeds chosen by greedy farthest-point sampling in the public chem-embedding cosine-distance space; this produces test sets that are diverse rather than locally similar. L5 trained on one library and tested on another in the same cell line, with no building-block-vocabulary overlap.

Features and models

Chemistry features were precomputed once as a 256-dim per-compound embedding shipped in `data/ZScreen_Canonical_Dataset/RNASeqAggregate/chem_embed.parquet`. The embedding is a fixed-seed Gaussian random projection from $\mathbb{R}^{2048}$ to $\mathbb{R}^{256}$ applied to per-compound 2048-bit ECFP4 fingerprints (Morgan radius 2), then L2-normalized. The projection matrix is held privately (NDA bundle) so that the embedding is not invertible to substructure information without it; under cosine similarity it preserves Tanimoto-like retrieval geometry approximately (Johnson-Lindenstrauss). Where building-block vocabularies overlapped between train and test, one-hot

encodings were fit on the training split only and combined with the chemistry embedding. Ridge regression with alpha equal to 10 was used as the primary prediction model. Ridge was chosen over a flexible nonlinear regressor because the goal of the ladder is to attribute generalization to the chemistry-to-phenotype function rather than to a model’s flexibility. Ridge alpha sweeps over four orders of magnitude were verified separately and do not change the qualitative ranking of rungs; the headline numbers are stable across the obvious hyperparameter envelope.

Baselines and metrics

Two baselines were used throughout. The first is a train-mean phenotype predictor (the constant prediction that minimizes squared error on the training set). The second is a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the 256-dim embedding space. The nearest-neighbor baseline is the most consequential one: in any chemistry split where the test molecules are close to training molecules, retrieval will appear strong, and reported model gains have to be evaluated against retrieval rather than against a trivial constant. Performance was summarized with row-wise cosine to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus nearest-neighbor retrieval, and per-compound win rate against retrieval.

Statistical reporting

Each random draw was evaluated with paired bootstrap resampling of test compounds. Confidence intervals were computed from 2,000 paired bootstrap samples, and summary tables report how many draws remained positive under the paired interval criterion. Ten random draws per rung were used rather than a single split because the size of test sets varies by holdout depth (some L3 splits drop to 67 compounds per draw in ZEL024) and a single random split would not give a stable estimate of model gain at small test-set sizes.

Cosine nearest-neighbor implementation

The chemistry-embedding nearest-neighbor baseline was implemented as an exact dot-product search on the L2-normalized 256-dim public embedding (cosine = dot product because the embedding is L2-normalized). Exact rather than approximate search was used because the same baseline appears in every cross-comparison in this manuscript and its public reproduction needs to be deterministic to the byte.

Limitations

The ladder is retrospective and strongest within selected library / cell-line systems rather than across every system in the canonical dataset. ZEL031 / THP1 supports the clearest extrapolation claim; ZEL024 / HEK293 is better interpreted as dense matrix completion. Cross-library scaffold-family transfer remains weak in the current data and should be treated as the next prospective experiment, not a solved capability. All transcriptomic targets are evaluated in the shared scVI latent space, so

full transcript-level validation and prospective measurement are needed before using the ladder as a final model-selection benchmark.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this working draft are organized in this repository under `paper3a/`, `paper2/tables/`, and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper3a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

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